SG No. 90 D_4^2 $P42_12$ [tetragonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `2a`, site symmetry: `2.22`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, 0, 0]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4,5,-6]`

Table 2: Wyckoff bond: `2b@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[0, 0, 0]$	`[1,-2,-7,8]`
2	$[X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4,-5,6]`

Table 3: Wyckoff bond: `2c@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,-7,-8]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,4,-5,-6]`

Table 4: Wyckoff bond: `4d@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, 0, 0]$	`[1,-8]`
2	$[-X, -X, Z]$	$[0, 0, 0]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-5]`

Table 5: Wyckoff bond: `4e@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[0, 0, 0]$	`[1,-7]`
2	$[-X, X, Z]$	$[0, 0, 0]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,-6]`

Table 6: Wyckoff bond: `4f@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,-6]`

continued ...

Table 6

No.	vector	center	mapping
4	[Y, X, 0]	[0, 0, 0]	[7,-8]

Table 7: Wyckoff bond: 8g@2a, bond symmetry: 1

No.	vector	center	mapping
1	[X, Y, Z]	[0, 0, 0]	[1]
2	[$-X$, $-Y$, Z]	[0, 0, 0]	[2]
3	[$-Y$, X, Z]	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3]
4	[Y, $-X$, Z]	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[4]
5	[$-X$, Y, $-Z$]	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[5]
6	[X, $-Y$, $-Z$]	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6]
7	[Y, X, $-Z$]	[0, 0, 0]	[7]
8	[$-Y$, $-X$, $-Z$]	[0, 0, 0]	[8]

* Wyckoff site: 2b, site symmetry: 2.22

Table 8: Wyckoff bond: 2a@2b, bond symmetry: 2.22

No.	vector	center	mapping
1	[X, X, 0]	$\left[0, 0, \frac{1}{2}\right]$	[1,-2,7,-8]
2	[$-X$, X, 0]	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,-4,5,-6]

Table 9: Wyckoff bond: 2b@2b, bond symmetry: 2.22

No.	vector	center	mapping
1	[X, $-X$, 0]	$\left[0, 0, \frac{1}{2}\right]$	[1,-2,-7,8]
2	[X, X, 0]	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,-4,-5,6]

Table 10: Wyckoff bond: 2c@2b, bond symmetry: 2.22

No.	vector	center	mapping
1	[0, 0, Z]	$\left[0, 0, \frac{1}{2}\right]$	[1,2,-7,-8]
2	[0, 0, Z]	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,-5,-6]

Table 11: Wyckoff bond: 4d@2b, bond symmetry: ..2

No.	vector	center	mapping
1	[X, X, Z]	$\left[0, 0, \frac{1}{2}\right]$	[1,-8]
2	[$-X$, $-X$, Z]	$\left[0, 0, \frac{1}{2}\right]$	[2,-7]
3	[$-X$, X, Z]	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,-6]
4	[X, $-X$, Z]	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,-5]

Table 12: Wyckoff bond: `4e@2b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-6]`

Table 13: Wyckoff bond: `4f@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-8]`

Table 14: Wyckoff bond: `8g@2b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `2c`, site symmetry: `4..`

Table 15: Wyckoff bond: `2a@2c`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2,3,4]`
2	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5,6,7,8]`

Table 16: Wyckoff bond: `4b@2c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7,-8]`

Table 17: Wyckoff bond: `8c@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, -z\right]$	`[8]`

* Wyckoff site: `4d`, site symmetry: `2..`

Table 18: Wyckoff bond: `4a@4d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, 0, -z]$	`[7,-8]`

Table 19: Wyckoff bond: `4b@4d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$[0, 0, -z]$	`[7,8]`

Table 20: Wyckoff bond: `8c@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, 0, -z]$	`[8]`

* Wyckoff site: `4e`, site symmetry: `..2`

Table 21: Wyckoff bond: `4a@4e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, x, 0]$	`[1,-7]`
2	$[-X, X, Z]$	$[-x, -x, 0]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, 0\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, 0\right]$	`[4,-6]`

Table 22: Wyckoff bond: `4b@4e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, x, 0]$	`[1,7]`
2	$[-X, -X, 0]$	$[-x, -x, 0]$	`[2,8]`
3	$[-X, X, 0]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, 0\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, 0\right]$	`[4,6]`

Table 23: Wyckoff bond: `8c@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, 0]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, 0\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$[x, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[-x, -x, 0]$	`[8]`

* Wyckoff site: `4f`, site symmetry: `..2`

Table 24: Wyckoff bond: `4a@4f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	`[4,-6]`

Table 25: Wyckoff bond: `4b@4f`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x, \frac{1}{2}\right]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[-x, -x, \frac{1}{2}\right]$	`[2,8]`
3	$[-X, X, 0]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	`[4,6]`

Table 26: Wyckoff bond: `8c@4f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-x, -x, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `8g`, site symmetry: `1`

Table 27: Wyckoff bond: `8a@8g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2} - y, x + \frac{1}{2}, z\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y + \frac{1}{2}, \frac{1}{2} - x, z\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, y + \frac{1}{2}, -z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, -z\right]$	`[6]`
7	$[Y, X, -Z]$	$[y, x, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[8]`